

Knowledge Graph Builder

Maharshi Soni | knowledge-graph | advanced

Description

Extracts entities and relationships from text using NLP, builds queryable knowledge graph with NetworkX. Shortest path, PageRank, connected components. FastAPI REST API.

Tech Stack

networkx, nltk, fastapi, uvicorn, pydantic, numpy, httpx, pytest

Features

- NLP entity extraction
- Relationship extraction
- NetworkX graph
- Graph queries: shortest path, PageRank
- FastAPI REST API
- Bulk ingestion

Sample Input

Text documents

```
curl -X POST /ingest -d '{"text": "John works at Google"}'
```

Sample Output

Graph query results

```
{"path": ["John", "works_at", "Google"], "entities": 45}
```

Project Files

```
.gitignore
LICENSE
project-idea.json
pytest.ini
README.md
requirements.txt
.github/workflows/test.yml
src/api.py
src/entity_extractor.py
src/graph_builder.py
```

```
src/knowledge_graph.py
src/main.py
src/models.py
src/nlp_utils.py
src/relationship_extractor.py
src/synthetic_data.py
src/__init__.py
tests/conftest.py
tests/test_api.py
tests/test_entity_extractor.py
tests/test_knowledge_graph.py
tests/test_relationship_extractor.py
tests/__init__.py
```

Test Results

PASSED

Run: 25 | Passed: 25 | Failed: 0

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